

Miao Wang

Professor

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Last Update: Aug. 2025

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EDUCATION

- **Tsinghua University** Beijing, China
Ph.D. in Computer Science and Technology, advisor: Prof. Shi-Min Hu Sept. 2011 – Jul. 2016
- **Xidian University** Xi'an, Shaanxi, China
Bachelor of Engineering in Computer Science and Technology (with honors) Sept. 2007 – Jul. 2011

WORK & RESEARCH EXPERIENCE

- **Beihang University** Beijing, China
Assistant Professor → Associate Professor → Professor Dec. 2018 – now
- **Tsinghua University** Beijing, China
Postdoc Researcher Jul. 2016 – Nov. 2018

RESEARCH INTERESTS

Virtual Reality, Mixed Reality, Computer Graphics, Visual Computing

SELECTED PUBLICATIONS (*CORRESPONDING AUTHOR)

1. **Miao Wang***, Wen-Tong Shu, Yi-Jun Li, Wanwan Li. Can I Get There? Negotiated User-to-User Teleportations in Social VR. *IEEE Transactions on Visualization and Computer Graphics (Special Issue of IEEE VR 2025)*, 31 (5), 2320-2330, 2025. (CCF-A) **IEEE VR 2025 Best Paper Award**
2. Jia-Qi Zhang, **Miao Wang***, Fu-Cheng Zhang, Fang-Lue Zhang. Skinned Motion Retargeting with Preservation of Body Part Relationships. *IEEE Transactions on Visualization and Computer Graphics*, Early Access, 2025. (CCF-A)
3. Yi-Jun Li, Hao-Zhong Yang, Wen-Tong Shu, **Miao Wang***. Semantics-Aware Avatar Locomotion Adaption for Indoor Cross-Scene AR Telepresence. *IEEE Transactions on Visualization and Computer Graphics*, Early Access, 2025. (CCF-A)
4. Weiqi Feng, Dong Han, Zekang Zhou, Shunkai Li, Xiaoqiang Liu, Pengfei Wan, Di ZHANG, **Miao Wang*** GPAvatar: High-fidelity Head Avatars by Learning Efficient Gaussian Projections. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 250-259, 2025. (CCF-A)
5. **Miao Wang***, Jinchao Zhou, Wei-Qi Feng, Yuzhu Jiang and Ana Serrano. Studying the Effect of Material and Geometry on Perceptual Outdoor Illumination. *IEEE Transactions on Visualization and Computer Graphics*, 30 (9), 6468-6480, 2024. (CCF-A)
6. **Miao Wang***, Yi-Jun Li, Jinchuan Shi and Frank Steinicke. SceneFusion: Room-Scale Environmental Fusion for Efficient Traveling Between Separate Virtual Environments. *IEEE Transactions on Visualization and Computer Graphics*, 30 (8), 4615-4630, 2024. (CCF-A)
7. **Miao Wang***, Ze-Yin Chen, Wen-Chuan Cai and Frank Steinicke. Transferable Virtual-Physical Environmental Alignment with Redirected Walking. *IEEE Transactions on Visualization and Computer Graphics*, 30 (3), 1696 - 1709, 2024. (CCF-A)
8. Hao-Bin Duan, **Miao Wang***, Yan-Xun Li, Yong-Liang Yang. Neural 3D Strokes: Creating Stylized 3D Scenes with Vectorized 3D Strokes. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 5240-5249, 2024. (CCF-A)
9. Jin-Chuan Shi, **Miao Wang***, Hao-Bin Duan, Shao-Hua Guan. Language Embedded 3D Gaussians for Open-Vocabulary Scene Understanding. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 5333-5343, 2024. (CCF-A)
10. Yi Zhang, Meng-Hao Guo, **Miao Wang**, Shi-Min Hu. Exploring Regional Clues in CLIP for Zero-Shot Semantic Segmentation. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 3270-3280, 2024. (CCF-A)

11. *Sen-Zhe Xu, Jia-Hong Liu, **Miao Wang**, Fang-Lue Zhang and Song-Hai Zhang* Multi-User Redirected Walking in Separate Physical Spaces for Online VR Scenarios. *IEEE Transactions on Visualization and Computer Graphics*, 30 (4), 1916 - 1926, 2024. **(CCF-A)**
12. *Jia-Qi Zhang, Hao-Bin Duan, Jun-Long Chen, Ariel Shamir and **Miao Wang**** HoughLaneNet: Lane Detection with Deep Hough Transform and Dynamic Convolution. *Computers & Graphics*, Volume 116, Pages 82-92, 2023. **CAD/Graphics 2023 Best Paper Award**
13. *Hao-Bin Duan, **Miao Wang***, Jinchuan Shi, Xuchuan Chen and Yan-Pei Cao* BakedAvatar: Baking Neural Fields for Real-Time Head Avatar Synthesis. *ACM Transactions on Graphics (Special Issue of SIGGRAPH Asia 2023)*, 42 (6), Article No. 225, 2023. **(CCF-A)**
14. *Jiaye Leng, Lili Wang*, Xiaolong Liu, Xuehuai Shi and **Miao Wang***. Efficient Flower Text Entry in Virtual Reality. *IEEE Transactions on Visualization and Computer Graphics*, 28 (11), 3662 - 3672, 2022. **(CCF-A)**
15. *Yi-Jun Li, Frank Steinicke and **Miao Wang****. A Comprehensive Review of Redirected Walking Techniques: Taxonomy, Methods, and Future Directions. *Journal of Computer Science and Technology*, 37 (3), 561-583, 2022. **(CCF-B)**
16. *Jun-Peng Xu, Chen-Yu Zuo, Fang-Lue Zhang and **Miao Wang****. Rendering-Aware HDR Environment Map Prediction From A Single Image. *Proceedings of the AAAI Conference on Artificial Intelligence*, 36 (3), 2857-2865, 2022. **(CCF-A)**
17. *Yi-Jun Li, Jinchuan Shi, Fang-Lue Zhang and **Miao Wang****. Bullet Comments for 360 Video. *Proceedings of IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR)*, 2022. **(CCF-A)**
18. *Jia-Qi Zhang, Xiang Xu, Zhi-Meng Shen, Ze-Huan Huang, Yang Zhao, Yan-Pei Cao, Pengfei Wan and **Miao Wang****. Write-An-Animation: High-level Text-based Animation Editing with Character-Scene Interaction. *Computer Graphics Forum (Special Issue of Pacific Graphics 2021)*, 40 (7), 217-228, 2021. **(CCF-B)**
19. *Ziming Ye, Junlong Chen, **Miao Wang*** and Yong-Liang Yang*. PAVAL: Position-Aware Virtual Agent Locomotion for Assisted Virtual Reality Navigation. *Proceedings of IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, 239-247, 2021. **(CCF-B)**
20. *Yi-Jun Li, **Miao Wang***, Frank Steinicke and Qinqing Zhao*. OpenRDW: A Redirected Walking Library and Benchmark with Multi-User, Learning-based Functionalities and State-of-the-art Algorithms. *Proceedings of IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, 21-30, 2021. **(CCF-B)**
21. *Ze-Yin Chen, Yi-Jun Li, **Miao Wang***, Frank Steinicke and Qinqing Zhao* A Reinforcement Learning Approach to Redirected Walking with Passive Haptic Feedback. *Proceedings of IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, 184-192, 2021. **(CCF-B)**
22. ***Miao Wang**, Ziming Ye, Jinchuan Shi and Yong-Liang Yang*. Scene-Context-Aware Indoor Object Selection and Movement in Virtual Reality. *Proceedings of IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR)*, 235-244, 2021. **(CCF-A)**
23. *Yi-Jun Li, De-Rong Jin, **Miao Wang***, Jun-Long Chen, Frank Steinicke, Shi-Min Hu and Qinqing Zhao*. Detection Thresholds with Joint Horizontal and Vertical Gains in Redirected Jumping. *Proceedings of IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR)*, 95-10, 2021. **(CCF-A)**
24. ***Miao Wang**, Xiao-Nan Fang, Guo-Wei Yang, Ariel Shamir and Shi-Min Hu*. Prominent Structures for Video Analysis and Editing. *IEEE Transactions on Visualization and Computer Graphics*, 27 (7), 3305-3317, 2021. **(CCF-A)**
25. *Xin Wen, **Miao Wang***, Christian Richardt, Ze-Yin Chen and Shi-Min Hu*. Photorealistic Audio-driven Video Portraits. *IEEE Transactions on Visualization and Computer Graphics (Special Issue of IEEE ISMAR 2020)*, 26 (12), 3457-3466, 2020. **(CCF-A)**
26. ***Miao Wang**, Yi-Jun Li, Wen-Xuan Zhang, Christian Richardt and Shi-Min Hu*. Transitioning360: Content-aware NFOV Virtual Camera Paths for 360 Video Playback. *Proceedings of IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, 241-250, 2020. **(CCF-B)**
27. ***Miao Wang**, Guo-Wei Yang, Shi-Min Hu, Shing-Tung Yau and Ariel Shamir*. Computational Video Montage from Themed Text. *ACM Transactions on Graphics (Proc. SIGGRAPH Asia)*, 38 (6), Article No.177, 2019. **(CCF-A)**
28. ***Miao Wang**, Guo-Ye Yang, Ruilong Li, Runze Liang, Song-Hai Zhang, Peter M. Hall and Shi-Min Hu*. Example-Guided Style-Consistent Image Synthesis from Semantic Labeling. *IEEE CVPR* 2019. **(CCF-A)**

TEACHING EXPERIENCE

- **Fundamental of Computer Graphics** Beihang University, Beijing
Instructor *Autumn 2022-Now*
- **Visual Media Computing** Beihang University, Beijing
Instructor *Spring 2023-Now*

ACADEMIC SERVICES

- **Conference Committee:**
 - **ISMAR 2019 - Beijing Night Co-Chairs**, Oct. 2019, Beijing, China
 - **ChinaVR 2024 - Program Co-Chairs**, Nov. 2024, Xiamen, China
- **Program Committee:**
 - **IEEE VR** 2021 - Now
 - **IEEE ISMAR** 2020 - Now
 - **Pacific Graphics** 2020 - 2022, 2024
 - **EuroGraphics** 2026
 - **Computational Visual Media** 2020-2025

SELECTED AWARDS

- **National Science and Technology Progress Award.** 2nd Prize, 9th Achiever, 2018
- **Young Elite Scientist Sponsorship Program by the China Association for Science and Technology.** 2020
- **Teaching Achievements of the Chinese Society of Image and Graphics.** Grand Prize, 2nd Achiever, 2024